

Business Taxation

(loosely follows Chapter 24 of Gruber)

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This Lecture vs. Next Lecture

- Next lecture will be from the perspective of the saver: How do taxes distort savings decisions and should we tax savings?
- This lecture is from the perspective of the investor: How do taxes affect investment / should we tax investment?
- Motivations:
 - ① Might care only about total investment/capital, not savers (e.g. if savers are all rich and government doesn't value their consumption, then Atkinson-Stiglitz result isn't main focus)
 - ② Savings doesn't have to equal investment in open economy

This Lecture's Big Questions

- ① If the government doesn't care about savers, should we tax capital?
- ② What are the theoretical and empirical investment effects of a business income tax?
- ③ What are the theoretical and empirical investment effects of a dividend income tax?
- ④ Is capital accumulation actually good for workers?
- ⑤ Are governments fighting a losing business tax battle?

Basic Definitions

- Business (firm) is a for-profit legal entity owned by shareholders typically with limited liability (if business goes bankrupt, share price drops to zero but shareholders not liable for unpaid bills/debt)
- Shareholders: Individuals who have purchased ownership stakes in a business.
- Ownership vs. control: owners are shareholders. Managers (CEO and top executives) in general do not own the company but run the corporation on behalf of shareholders

- Firms can finance themselves through debt or through equity
- Equity finance: The raising of funds by sale of ownership shares in a firm. Shareholders receive dividends from corporation and capital gain if the share price increases
- Debt finance: The raising of funds by borrowing from lenders such as banks, or by selling corporate bonds. Corporate bonds are promises by a business to make periodic interest payments, as well as ultimate repayment of principal, to the bondholders (the lenders)
- Bond holders have priority over shareholders for repayment in case of bankruptcy

- Businesses use capital (land, buildings, machines, equipment) and labor (workers) to transform inputs (raw materials) into outputs (goods produced and sold).
- Profits = revenues from goods sold - expenses (labor costs, inputs, capital depreciation, interest payments on debt)
- Profits are double taxed:
 - ① Business income tax (τ^{INC}): Tax on annual profits (e.g. the corporate income tax of 21%)
 - ② Payout tax: Tax on after-business-income-tax profits once they are distributed (paid out) to shareholders

- Payout tax at the shareholder level happens via either of two similar forms:
 - ① Dividend: Standard payout form. A specific amount is paid per share owned, often in regular amounts. Owners pay dividend tax (τ^{DIV})
 - ② Capital gain: If business keeps after-business-income-tax profits, share value rises. Owners realize capital gain when selling those shares, pay capital gains tax

Why Not Tax Payouts Only?

- Businesses are not people but are ultimately owned by people. Starting point is usually to tax people based on their economic resources but:
 - ① Taxing Pure Profits: Some firms have market power (e.g., Microsoft) and hence earn pure profits. Taxing pure profits does not distort behavior because firms maximize profits anyway
 - ② Back-up for individual taxes: If corporations were not taxed on their earnings, then individuals who owned shares in corporations could postpone taxes indefinitely by having the corporations never pay out their earnings
 - ③ Taxing foreign owners: Corporations often have foreign owners. Countries want to tax economic activity on their territory. E.g., consider developing country with foreign-owned mineral/oil extraction companies
 - ④ Tax collection convenience: Historically, corporations are more convenient to tax than individuals because they are large, visible, and have detailed accounts (for transparency for their shareholders)

Should We Tax Capital if We Don't Care about Savers?

- Mankiw (2000) and Judd (1985): Suppose you only want to maximize the welfare of “spenders” (workers who consume all their income every year and save nothing) and don't care about “savers”.
- In standard model where government must fund a base spending level (e.g. an army):
 - Substitution effect of capital tax on savings dominates income effect
 - Savings = Investment
 - Firms do not earn pure profits
 - Capital augments labor
- Spenders (in the long run) would want a zero capital tax and a positive labor income tax!
- Mechanism: Capital augments labor (makes labor so much more productive and thus increases workers' wages) that any capital stock reduction via a capital tax is not worth it

But These are Empirical Questions

- Do capital taxes really reduce investment?
- Does capital really augment labor?

- Classic cost-of-capital model (Hall-Jorgenson 1967):
- Firm has concave gross-profit function $F(K)$ (equal to revenue minus input and labor costs), has to pay capital cost $r \cdot K$ (cost of buying or leasing machines/buildings).
- Firm maximizes:

$$\max_K F(K) - r \cdot K$$

$$FOC : F'(K) = r$$

- If $F(K^*) - rK^* > 0$, we say firm earns pure profit.
- If $F(K^*) - rK^* = 0$, firm does not earn pure profit (and thus barely breaks even).

- Now firm has to pay a business income tax on its gross profit $F(K)$.
- Firm maximizes:

$$\max_K (1 - \tau^{INC}) \cdot F(K) - r \cdot K$$

$$FOC : F'(K) = \frac{r}{(1 - \tau^{INC})}$$

- $\Rightarrow$ Business income tax reduces investment

- But what if the business income tax is assessed on profits after capital costs?
- Firm maximizes:

$$\max_K (1 - \tau^{INC}) \cdot [F(K) - r \cdot K]$$

$$FOC : F'(K) = r$$

- $\Rightarrow$ Business income tax has no effect on investment! (Robertson 1927)

What Just Happened Here?

- When all costs are deducted from taxable income, the business income tax is a tax on pure profit
- The K that maximizes pure profit $F(K) - r \cdot K$ also maximizes 79% of pure profit
- Firms in perfectly competitive markets do not earn pure profit. Best they can do is break even (after capital costs).
- How competitive are markets?
- Introspection test: Would you care if you were banned from buying from your favorite grocery store, airline, car company, restaurant, etc.?
- Empirical test: Many markets are quite competitive, many are not

Are Business Costs Fully Deductible?

- Most costs are fully and immediately deductible.
- “Deductible” means “you can deduct the amount from your taxable income (in this case, from your gross profits)”
- Capital costs (purchases of machines/buildings) cannot be deducted immediately and instead must be made in pre-defined sequence of annual “depreciation deductions”
- Deductions add up to original nominal cost, but total effective deductibility is only partial because value of future tax deductions must be discounted
- Longer depreciation schedules $\Rightarrow$ less effective deductibility
- Ex. Cars depreciated over 5 years, railroads over 20 years

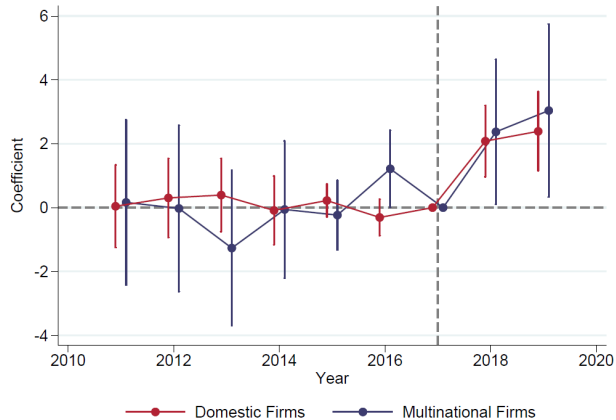
Evidence on Business Income Taxes and Investment

- Old literature trying to estimate elasticity of business investment with respect to $(1 - \tau^{INC})$ and other changes in cost of capital: the pre-tax rate of return required on marginal investments
- Hassett-Hubbard (2002) consensus range of cost-of-capital elasticity of investment: $[-0.5, -1.0]$ (very elastic)
- Some of that evidence is based on (questionable) analysis of changes in τ^{INC}
- Would imply that business income tax is very far from pure profits tax

New Evidence on Business Income Taxes and Investment

- 2017 tax cut reduced the cost of capital for some firms more than others
- Large firms (due to a special provision called Section 179) and firms with slowly depreciating capital experienced a larger cost of capital reduction than others
- Chodorow-Reich, Smith, Zidar, and Zwick (2025) find that those firms increased investment more after 2017

A. Domestic Cost-of-Capital $\hat{\Gamma}$



Source: Chodorow-Reich, Smith, Zidar, and Zwick (2025)

Business Income Tax Theory vs. Practice

- Theory recommends: *narrow* the business income tax base (allow for more capital cost deductions) and *increase* the business income tax rate
- Neither party has followed this recommendation
- “President’s Framework for Business Tax Reform” (Obama 2012): Lengthen depreciation schedules and reduce corporate income tax rate from 35% to 28%
- Trump tax cut: Reduced rate from 35% to 21%, kept depreciation schedules short

Why the Disconnect?

- Possibility 1: Interest deductibility means we actually subsidize investment, so narrowing the base makes the tax code neutral (but then why lower the rate?)
- Possibility 2: International competition (discussed later)
- Possibility 3: Ideal solution cannot be fully enforced
 - Example: A firm with \$10m in pure profit buys a \$10m airplane to erase taxable income, then owners effectively take private control of the plane
 - In principle this should be taxed as ordinary income, but enforcement is difficult
 - Full deductibility can erode the tax base substantially
- Possibility 4: Policymakers are wrong (many think so)

There is no equity-efficiency tradeoff to business taxes. The government can raise revenue for redistribution with no negative impacts on the economy.

- A. True
- B. False
- C. Uncertain

Dividend Tax Theory

- Dividend taxes: second layer of double-taxation of business profit.
- Applies to after-annual-business-income-tax profits, once paid to shareholders (owners).
- Two views of dividend taxation: “traditional view” and “new view”
 - “Traditional view”: Dividend tax is just like the annual business income tax
 - “New view”: Dividend tax is different
- Key: Source of investment funds

Traditional View of Dividend Taxes

- Leading treatment: Poterba-Summers (1985). Same as Hall-Jorgenson, but will write in slightly different way.
- Think of an entrepreneur (owner of new firm) maximizing her after-tax wealth
- Entrepreneur starts with wealth W and decides how much of that wealth to use for her firm's investment: i.e. how much equity to contribute so the firm can buy capital K
- Firm shuts down after one year and profits paid as dividends
- Whatever she doesn't use for investment is kept in a bank and earns r

Traditional View of Dividend Taxes

- Entrepreneur chooses K to maximize her after-tax return on wealth:

$$\max_K [(1 - \tau^{DIV}) \cdot (1 - \tau^{INC}) \cdot F(K) + r \cdot (W - K)]$$

$$FOC : F'(K) = \frac{r}{(1 - \tau^{DIV}) \cdot (1 - \tau^{INC})}$$

- $\Rightarrow$ Dividend tax reduces investment, in exactly the same way as the annual business income tax

New View of Dividend Taxes

- Developers: King (1977), Auerbach (1979), Bradford (1981)
- Think of owner of an already profitable firm (e.g. Microsoft)
- Firm starts with profits from preexisting operations *PPO* that are enough to pay for any profitable amount of investment
- Owner decides how much of profits from preexisting operations to use for her firm's investment
- Firm shuts down after one year and profits paid as dividends
- Whatever she doesn't use for investment is paid out as a dividend and earns r in the bank

New View of Dividend Taxes

- Owner chooses K to maximize her after-tax return on wealth:

$$\max_K [(1 - \tau^{DIV}) \cdot (1 - \tau^{INC}) \cdot F(K) + r \cdot (1 - \tau^{DIV}) \cdot (PPO - K)]$$

$$FOC : F'(K) = \frac{r}{(1 - \tau^{INC})}$$

- $\Rightarrow$ Dividend tax does not affect investment!
- To see what's going on:

$$FOC : (1 - \tau^{DIV}) \cdot (1 - \tau^{INC}) \cdot F'(K) = r \cdot (1 - \tau^{DIV})$$

Traditional View vs. New View

- Key: What is the opportunity cost of investment (i.e. what is the cost of capital)?
- Traditional view: Owner forgoes $\$r$ for every \$1 of investment
- New view: Owner forgoes only $\$r \cdot (1 - \tau^{DIV})$ for every \$1 of investment
- Why? Because there are available funds (profits from preexisting operations) that will be subject to dividend taxes regardless of whether they're used for investment

Traditional View vs. New View

- Thus in traditional view: dividend tax does not affect the opportunity cost of investment (r) but reduces the post-tax return on investment by $(1 - \tau^{DIV})$
- But in new view: dividend tax reduces the opportunity cost of investment by the exact same factor $(1 - \tau^{DIV})$ that it reduces the post-tax return on investment
- In new view, dividend tax is effectively a tax on past investment, not new investment

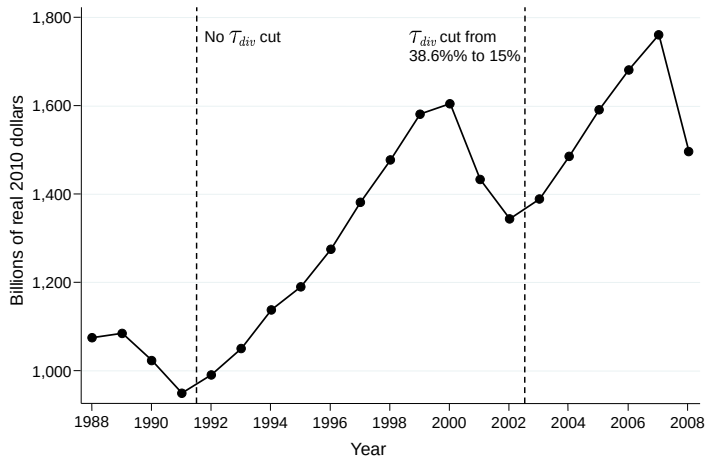
Testing between the Two Views

- Facebook in 2005: Clearly a traditional-view firm (no preexisting operations)
- Microsoft now: Clearly a new-view firm (huge profits from preexisting operations)
- What about the average firm? Hard to say from just looking at investment data
- Need variation in dividend tax rates

The 2003 Dividend Tax Cut

- Reduced top federal dividend tax rate from 38.6% to 15%
- Traditional-view argument from President: provide “near-term support to investment” and “capital to build factories, to buy equipment, hire more people”
- Want to test whether business investment rose after 2003
- Problem: Must control for business cycle

U.S. Corporate Investment NIPA Private Fixed Non-residential Investment



Source: Yagan (2015)

DD Design Around 2003 Dividend Tax Cut

- Yagan (2015): Use unaffected corporations to control for the business cycle
- C-corporations: Traditional corporation type, subject to double taxation (annual business income tax and dividend tax)
- S-corporations: “Pass-through” business type subject only to annual business income tax (assessed at owner level), never subject to dividend tax
- DD between C-corporations (directly affected by 2003 dividend tax cut) and S-corporations (not directly affected)

Strategy: C-corporations vs. S-corporations

- After incorporating, a corporation elects either C or S tax status

	Tax rate on annual income	Tax rate on dividends
C-corporations (treatment)	35%	15%
S-corporations (control)	35%	0%

- S-corporations: <100 non-institutional shareholders, one stock class
- Compete in the same narrow industries and at the same scale throughout the United States → common trends

Source: Yagan (2015)

Example: Retail hardware chains



- Largest hardware chain
- C-corporation



- Third-largest hardware chain
- S-corporation

Note: Only public sources were used to populate this slide.

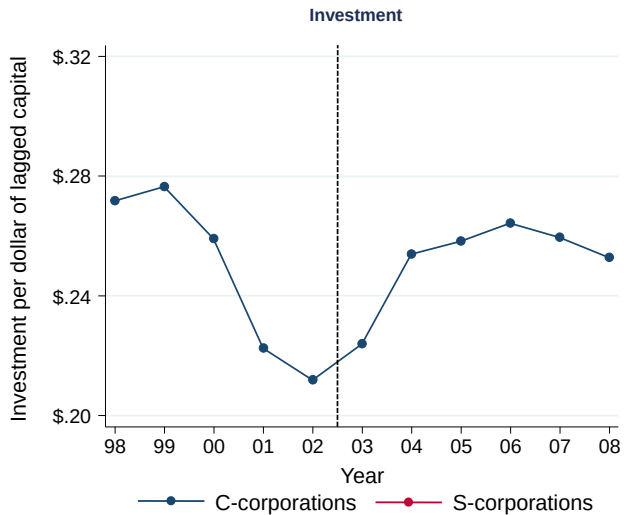
Source: Yagan (2015)

Example: Retail Hardware Chains in Suburban Chicago

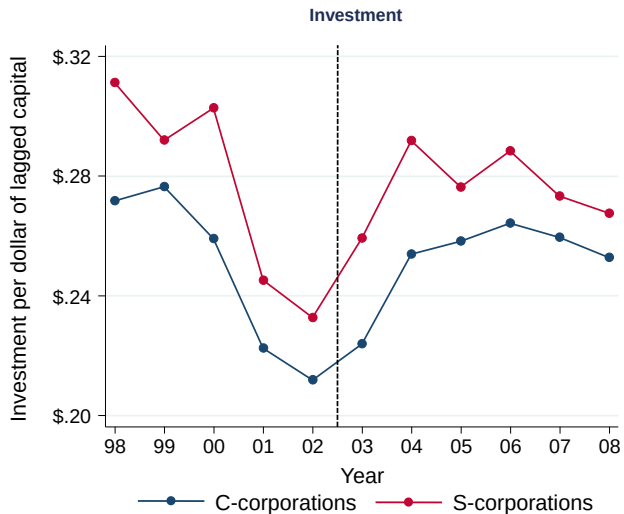


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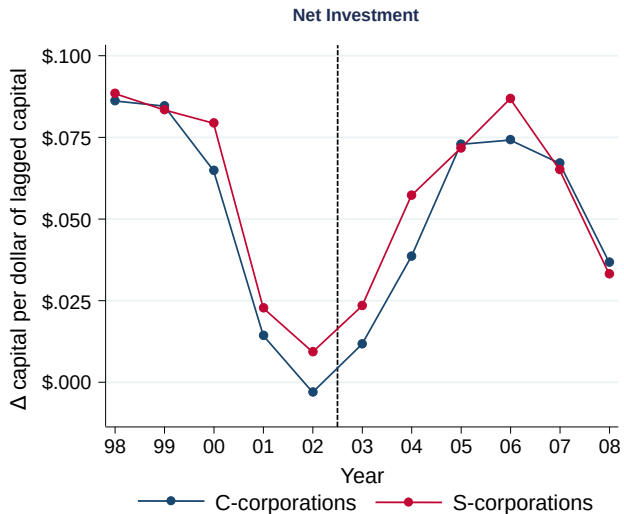
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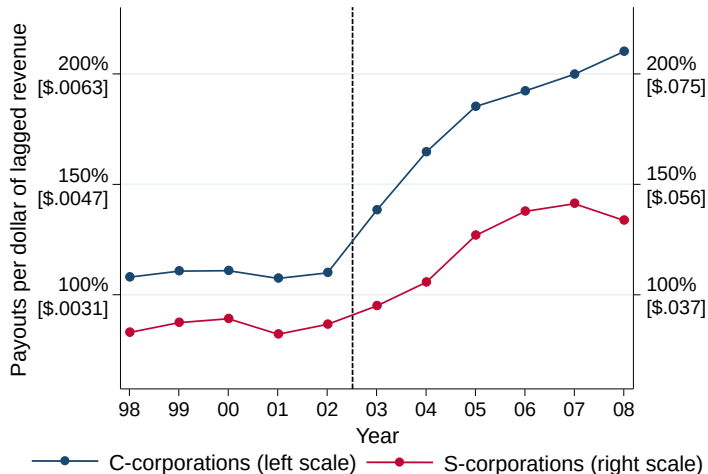


Source: Yagan (2015)



Source: Yagan (2015)

Payouts to Shareholders



Source: Yagan (2015)

- Elasticity of investment wrt $(1 - \tau_{DIV}) = 0.00$
- 95% confidence upper bound: 0.08
- Traditional view prediction: $[0.21, 0.41]$, depending on cost-of-capital elasticity of investment (Hassett-Hubbard consensus range $[-0.5, -1.0]$)

- Natural interpretation: New view is mostly right
- Caveat #1: Traditional view could be right, but Hassett-Hubbard are wrong and investment barely responds to cost-of-capital changes (i.e. $F(K)$ very concave)
- Caveat #2: Traditional view could be right if tax/political system didn't get in the way (e.g. gov cannot commit to permanently low dividend rate)
- Regardless: Future dividend tax changes unlikely to affect investment

Is Capital Accumulation Good for Workers?

- Central assumption in all these models: Capital complements labor $\Rightarrow$ investment is good for workers
- Strong reason to think this has been true in long arc of history
- Now/future: Capital may complement high-skilled (e.g. computers) but not low-skilled
- Intuition: You don't need more than one car.
- Formally: Products produced by low-skilled workers may have hit inelastic demand $\Rightarrow$ More efficient production no longer increases quantity sold $\Rightarrow$ Need fewer and fewer low-skilled workers making cars

Big Mac Index

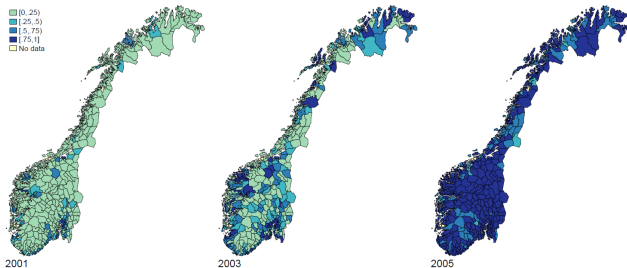
Table 1: McDonalds Cashier or Crew Wages and Big Mac Prices, December 1998

<i>Country</i>	Estimated hourly wage rate	Reported Big Mac price	Exchange Rate per \$1	\$ hourly wage rate	\$ Big Mac price	Economist \$ Big Mac 3/99**	Big Macs per hour of work
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Russia	8.00	25.5	19.57	0.41	1.30		0.31
Korea	1700	3000	1210	1.41	2.48		0.57
Brazil	2.87	4.45	1.73*	1.66	2.57	1.71	0.65
Poland	4.12	5.3	3.50	1.18	1.51	1.38	0.78
Czech Rep.	45.00	53	30.30	1.49	1.75		0.85
UK	3.60		0.62*	5.80	3.07	3.07	1.89
USA	6.00		1.00	6.00	2.43	2.43	2.12
Germany	11.28	4.95	1.67	6.76	2.97	2.72	2.28
France	40.22	17.5	5.76	6.99	3.04	2.87	2.30
Italy	10417	4500	1646	6.33	2.73	2.5	2.31
Belgium	280.00	114	34.50	8.12	3.30		2.46
Sweden	64.90	25	8.03	8.09	3.11	2.88	2.60
Japan	844	280	120*	7.03	2.33	2.44	3.01

Ashenfelter-Jurajda (2001)

Case study of complementarity: Rollout of broadband

Figure A4. Geographical distribution of broadband coverage rates.

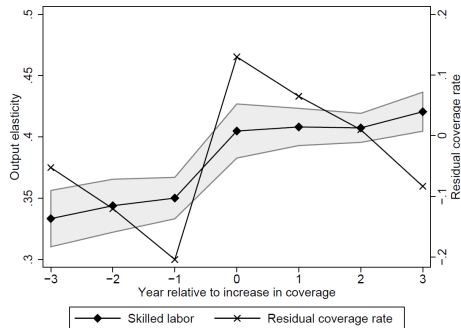


Note: The graphs show the geographical distribution of broadband coverage rates of households in 2001, 2003 and 2005.

Source: Akerman-Gaarder-Mogstad (2013)

Case study of complementarity: Rollout of broadband

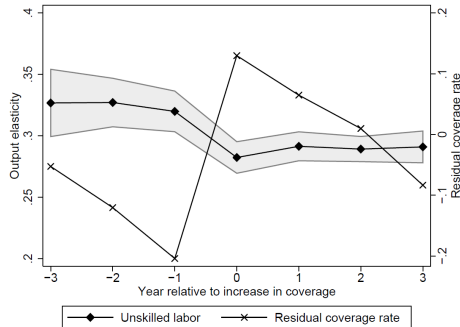
(a) Output elasticity: Skilled labor



Source: Akerman-Gaarder-Mogstad (2013)

Case study of complementarity: Rollout of broadband

(b) Output elasticity: Unskilled labor



Source: Akerman-Gaarder-Mogstad (2013)

Are Governments Fighting a Losing Business Tax Battle?

- Firms are very good at minimizing taxes
- Multinational firms avoid taxes through complex international relationships
- Domestic firms avoid taxes by strategic use of entity classification

Taxation of Multinational Firms

- Multinational firms: Firms that operate in multiple countries.
- Subsidiaries: The production arms of a corporation that are located in other nations.
- Territorial tax system: Corporations earning income abroad pay tax only to the government of the country in which the income is earned (most countries use this system)
- Global tax system: Corporations are taxed by their home countries on their income regardless of where it is earned (US system, foreign profits used to be taxed only when “repatriated”, now taxed immediately at lower rate)
- Foreign tax credit: U.S.-based multinational corporations may claim a credit against their U.S. taxes for any tax payments made to foreign governments when funds are repatriated to the US parent.

Tax Avoidance of Multinationals (Zucman '14)

- Share of profits made abroad by US corporations is about 1/3 today (was less than 5% in the 1930s)
- 50% of foreign profits of multinationals are reported in tax havens (such as Ireland)
- Multinational companies are particularly savvy to avoid corporate income tax by reporting most of their profits in low tax countries using transfer pricing: one subsidiary buys/sells to another at manipulated prices to transfer profits
- $\Rightarrow$ 20% of profits of US corporations is retained in tax havens
- $\Rightarrow$ Effective corporate tax rate is lower than nominal US Federal tax rate

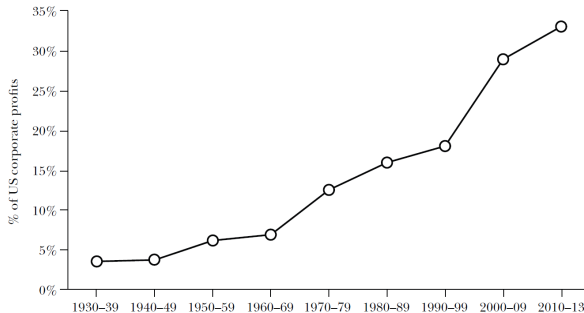
Ex: Microsoft's Irish sub Round Island Inc.



- In 2005: 1.5% of employees, 23% of assets
- Microsoft 2004 average tax rate: 33%
- Microsoft 2005 average tax rate: 26% due to “foreign earnings taxed at lower rates”

Rising importance of earnings booked abroad

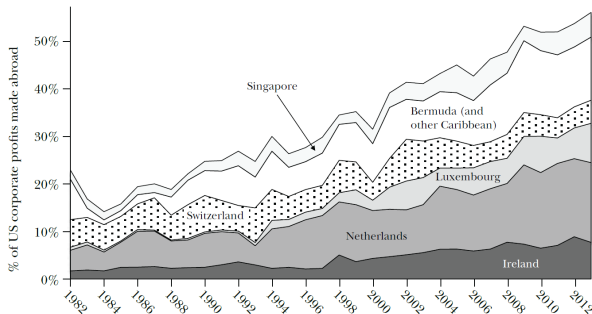
The Share of Profits Made Abroad in US Corporate Profits



Source: Zucman (2014)

Rising importance of earnings booked abroad

The Share of Tax Havens in US Corporate Profits Made Abroad



Source: Zucman (2014)

- Other way for U.S. corporations to dodge U.S. corporate tax: change country of incorporation to a tax haven
- Cannot just say “I’m an Irish corporation now.” Must merge with an Irish corporation first, called “corporate inversion”
- Ex. Medtronic (maker of heart pacemakers) merged with Irish Covidien in 2014 → Declared legal headquarters in Ireland → Avoided U.S. tax on \$14bn held overseas
- Potential rationale for low U.S. corporate tax rate: Corporations will move headquarters/jobs overseas
- No evidence though yet that many actual jobs move (e.g. Medtronic kept operational HQ in Minnesota)

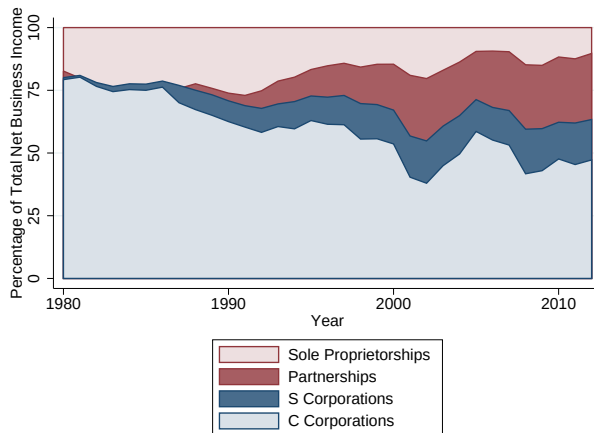
Domestic Firms Avoiding Taxes

- Businesses taxed as either C-corporations or pass-throughs
- Between 1986-2017, there was a clear tax advantage for companies to be pass-throughs and for owner-managers to report their income as profit rather than wages
- Major pass-through types: S-corporations and partnerships
- Evidence: Cooper-McClelland-Pearce-Prisinzano-Sullivan-Yagan-Zidar-Zwick (2016), Smith-Yagan-Zidar-Zwick (2019)

Domestic Firms Avoiding Taxes: Tax-Rate Comparison

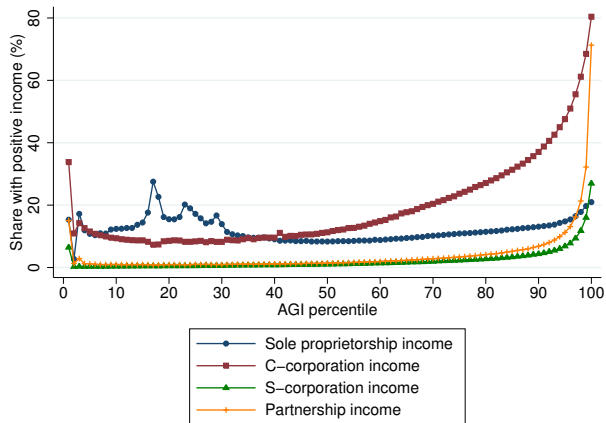
- C-corporate profit taxation (2003–2017): annual corporate income tax (35%) + dividend tax (24%)
- Pass-through profit taxation (2003–2017): annual income taxed at owner level (max 39.6%) + no dividend tax
- Pass-through wage taxation (2003–2017): annual income taxed at owner level (max 39.6%) + Medicare/ObamaCare surtax (max 3.8%)
- Implication: strong incentive to classify labor income as pass-through profit

THE RISE OF PASS-THROUGHS



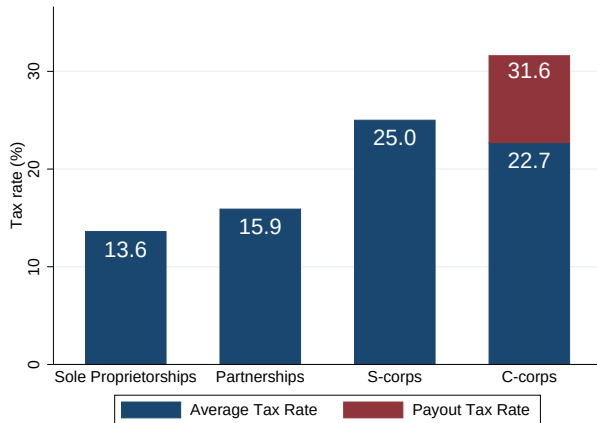
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BUSINESS PARTICIPATION RATES BY AGI PCTILE

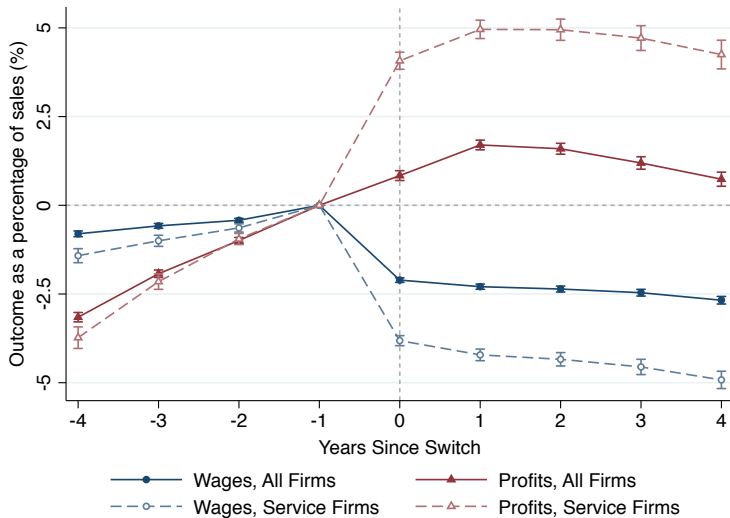


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TAX RATE BY ENTITY TYPE



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Proposals to Tax Business More Effectively

- Multinationals: Apportion global profits based on sales location
- Ex. Microsoft 2005 profit = \$12bn and 67% of sales were to U.S. customers $\Rightarrow$ \$8bn would be subject to U.S. tax
- Makes Irish subsidiary/inversion irrelevant
- Customers will not all move to Ireland
- This is exactly what U.S. states do

Proposals to Tax Business More Effectively

- Domestic businesses: Put size limit on pass-throughs
- Difference between labor and capital income is fuzzy for small businesses $\Rightarrow$ Good reason to tax both identically at owner level, even if small businesses taxed less than big businesses
- Little good reason to allow big businesses to be taxed at different rates (e.g. Home Depot vs. Menards)

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